Teaching demo

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Uniform and Pointwise Convergence for Sequences of Functions on $\mathcal{C}[0,1]$

We look at two notions of convergence for sequences of functions. The stronger one — uniform convergence — preserves continuity of functions.

What I assume students have seen

- Sequences of real numbers and their convergence
  (Reminder: we say that $(a_n)_{n\in\mathbb{N}}$ converges to L if for all $\varepsilon > 0$, there exists $n_0 \in \mathbb{N}$ s.t. $|a_n - L| < \varepsilon$ for all $n \geq n_0$)
- The definition of a sequence of functions and some examples.

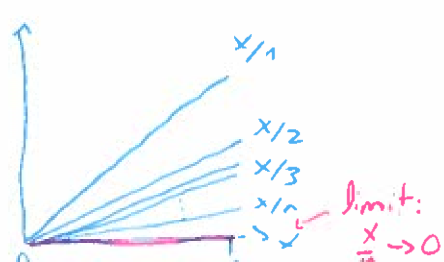
(Definition

Let $D \subseteq \mathbb{R}$ and, for each n in $\mathbb{N}$, let f_n be a function from D into $\mathbb{R}$. Then, $(f_n)_{n\in\mathbb{N}}$ is a sequence of functions on D. For each $x \in D$, the sequence $(f_n(x))_{n\in\mathbb{N}}$ is a sequence of real numbers.

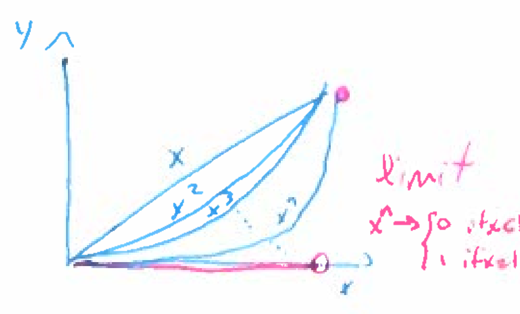
Remark: Two types of variables: $n \in \mathbb{N}$, $x \in \mathbb{R}$

Examples

$\left(\frac{x}{n}\right)_{n\in\mathbb{N}}$



$(x^n)_{n\in\mathbb{N}}$



Convergence

Definition

If, for every $x \in D$, $(\underbrace{f_n(x)}_{\substack{\text{real number} \\ \text{for fixed } x}})_{n\in\mathbb{N}}$ converges, then the sequence of functions $(f_n)_{n\in\mathbb{N}}$ converges pointwise.

Its limit, $F(x) = \lim_{n\to\infty} f_n(x)$, is the limit function

Remark: The limit function satisfies:

"For all $\varepsilon > 0$ and for all $x \in D$, there exists $n_0 \in \mathbb{N}$ s.t.

$|f_n(x) - F(x)| < \varepsilon \quad \forall\, n \geq n_0$."

Example

$(x/n)_{n\in\mathbb{N}}$.

For a fixed $x \in [0,1]$, $0 \leq \frac{x}{n} \leq \frac{1}{n}$

By the squeeze theorem, $\frac{x}{n} \to 0$ as $n \to \infty$.

Example

$(x^n)_{n\in\mathbb{N}}$

- For $x \in [0,1)$, $x^n \to 0$ as $n \to \infty$
- For $x = 1$, $x^n = 1^n \to 1$ as $n \to \infty$

$(x^n)_{n\in\mathbb{N}}$ converges pointwise to $F(x) = \begin{cases} 0 & \text{if } x \in [0,1) \\ 1 & \text{if } x = 1 \end{cases}$

Problem: • x^n is continuous on $[0,1]$ for all $n \in \mathbb{N}$

• $\lim_{n\to\infty} x^n$ is not continuous on $[0,1]$

Pointwise convergence does not preserve continuity...

Uniform convergence

Definition

Given a sequence of functions $(f_n)_{n\in\mathbb{N}}$ on D and a function F on D, $(f_n)_{n\in\mathbb{N}}$ converges uniformly to F if, for all $\varepsilon>0$, there exists $n_0 \in \mathbb{N}$ such that

$$|f_n(x) - F(x)| < \varepsilon \quad \text{for all } n > n_0 \text{ and all } x \in D$$

Remark: n_0 depends only on ε, not on x.

Why we care? Uniform convergence preserves continuity (Theorem, coming later)

Example

$\left(\frac{x}{n}\right)_{n\in\mathbb{N}}$ converges uniformly to 0 on $D=[0,1]$

Proof Let $\varepsilon>0$. We know $\left|\frac{x}{n}\right| < \frac{1}{n}$. Let $n_0 \in \mathbb{N}$ be s.t. $\frac{1}{n_0} < \varepsilon$ (exists by Archimedean Principle). Then, if $x \in [0,1]$ and $n \geq n_0$,

$$\left|\frac{x}{n} - 0\right| \leq \frac{1}{n} \leq \frac{1}{n_0} < \varepsilon.$$

So $\left(\frac{x}{n}\right)_{n\in\mathbb{N}}$ converges uniformly

Example

$(x^n)_{n\in\mathbb{N}}$ does not converge uniformly on $[0,1]$.

Proof idea

We show that it converges uniformly on $[0,c]$, for any $0<c<1$, and we highlight what part of the argument fails for $c=1$.

- $(x^n)_{n\in\mathbb{N}}$ converges uniformly to 0 on $[0,c]$, for $0<c<1$.

Let $\varepsilon>0$. We want to find n_0 such that $|x^n-0|<\varepsilon$ for $n\geq n_0$.

("Draft": $|x^n-0| = x^n \leq c^n < \varepsilon \iff n = \dfrac{\ln(\varepsilon)}{\ln(c)}$. Since $x^n \leq x^{n_0}$ if $n\geq n_0$, we let $n_0 \geq \ln(\varepsilon)/\ln(c)$).

Let $n_0 \geq \log(\varepsilon)/\log(c)$. Then, for $n\geq n_0$,

$$|x^n-0| = x^n \leq x^{n_0} \leq c^{n_0} \leq c^{\ln(\varepsilon)/\ln(c)} = \varepsilon.$$

So $(x^n)_{n\in\mathbb{N}}$ converges uniformly to 0 on $[0,c]$.

- Why it does not work when $c=1$?

If $c=1$, $\ln(c)=\ln(1)=0$, so $\dfrac{\ln(\varepsilon)}{\ln(c)}$ is undefined.

As $c\to 1$, n_0 has to be larger and larger. $\boxtimes$

Theorem

If $f_n \to F$ uniformly and f_n is continuous for all $n\in\mathbb{N}$, then F is continuous.

Proof: next class...